THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL2108: PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT

Dec 9, 2022 2:30 PM - 5:30 PM

Answer ALL questions

A compound interest table is provided.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

1. The university is planning to build a new hall to accommodate more students, they are considering integrating the concept of circular economy in the design, construction, and operation of this hall. As the engineering consultant of this project, please explain to the university:
   a) The concept of circular economy (5 marks)
   b) Is a circular building sustainable? (5 marks)
   c) Recommend circular strategies that could be adopted in this hall and give examples for each strategy. (15 marks)

2. There are four alternative ways to renovate a hotel. The capital costs needed, the extra annual OMR costs required, the extra annual benefit generated from the renovation are as follows:

	Alternative 1 $\$ \times 10^3$	Alternative 2 $\$ \times 10^3$	Alternative 3 $\$ \times 10^3$	Alternative 4 $\$ \times 10^3$
Capital of the renovation	150	185	250	400
Extra annual OMR costs needed	15	20	25	35
Extra annual benefit generated	50	60	80	110

   Assuming the life of the project to be 10 years and the discount rate to be 10% per annum, find the alternative which gives:
   a) The highest net annual benefit [8 marks]
   b) The highest net present worth [8 marks]
   c) The highest direct B/C ratio [4 marks]
   d) Using the incremental benefit-cost ratio method to find the best alternative [5 marks]

3. The following table shows the duration and the logical relationship of the activities involved in a construction project:

Activity ID	Predecessor	Duration (days)
A	-	12
B	A	10
C	A	20
D	B	15
E	C	30
F	C	18

G	D, E	26
H	F, G	14

a) Prepare an activity-on-node network diagram based on the information given in the table above and identify the critical path of this project. (10 marks)
b) Prepare a table showing the early and late start and completion dates of each activity and the free floats (10 marks)

4. The Cross Bay Link in Tseung Kwan O is set to open in Dec 2022. The project consists of an about 1.8 kilometers long dual two-lane carriageway with cycle track and footpath. Upon completion, the Cross Bay Link will relieve the traffic congestion within the Tseung Kwan O town center and along Wan Po Road.

a) Please identify the social benefits and social costs of the project (15 marks)
b) List participants who are likely to be involved in the following stages of the project, describe their roles and responsibilities:
   i. conceptual planning and feasibility study (5 marks)
   ii. procurement (5 marks)
   iii. construction (5 marks)

Compound Interest Table

$i = 10\%$

n	caf $(1+i)^n$	pwf $\frac{1}{(1+i)^n}$	uscaf $\frac{(1+i)^n-1}{i}$	ussff $\frac{i}{(1+i)^n-1}$	uspwf $\frac{(1+i)^n-1}{i(1+i)^n}$	uscrf $\frac{i(1+i)^n}{(1+i)^n-1}$
1	1.1000	0.9091	1.0000	1.0000	0.9091	1.1000
2	1.2100	0.8264	2.1000	0.4762	1.7355	0.5762
3	1.3310	0.7513	3.3100	0.3021	2.4869	0.4021
4	1.4641	0.6830	4.6410	0.2155	3.1699	0.3155
5	1.6105	0.6209	6.1051	0.1638	3.7908	0.2638
6	1.7716	0.5645	7.7156	0.1296	4.3553	0.2296
7	1.9487	0.5132	9.4872	0.1054	4.8684	0.2054
8	2.1436	0.4665	11.436	0.0874	5.3349	0.1874
9	2.3579	0.4241	13.579	0.0736	5.7590	0.1736
10	2.5937	0.3855	15.937	0.0627	6.1446	0.1627
11	2.8531	0.3505	18.531	0.0540	6.4951	0.1540
12	3.1384	0.3186	21.384	0.0468	6.8137	0.1468
13	3.4523	0.2897	24.523	0.0408	7.1034	0.1408
14	3.7975	0.2633	27.975	0.0357	7.3667	0.1357
15	4.1772	0.2394	31.772	0.0315	7.6061	0.1315
16	4.5950	0.2176	35.950	0.0278	7.8237	0.1278
17	5.0545	0.1978	40.545	0.0247	8.0216	0.1247
18	5.5599	0.1799	45.599	0.0219	8.2014	0.1219
19	6.1159	0.1635	51.159	0.0195	8.3649	0.1195
20	6.7275	0.1486	57.275	0.0175	8.5136	0.1175
21	7.4002	0.1351	64.002	0.0156	8.6487	0.1156
22	8.1403	0.1228	71.403	0.0140	8.7715	0.1140
23	8.9543	0.1117	79.543	0.0126	8.8832	0.1126
24	9.8497	0.1015	88.497	0.0113	8.9847	0.1113
25	10.835	0.0923	98.347	0.0102	9.0770	0.1102
26	11.918	0.0839	109.18	0.0092	9.1609	0.1092
27	13.110	0.0763	121.10	0.0083	9.2372	0.1083
28	14.421	0.0693	134.21	0.0075	9.3066	0.1075
29	15.863	0.0630	148.63	0.0067	9.3696	0.1067
30	17.449	0.0573	164.49	0.0061	9.4269	0.1061

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